NAME: Mary Nyambura Gitau
GMAIL:marynyambura657@gmail.com
ADMISSION NUMBER: CS-SA12-26067

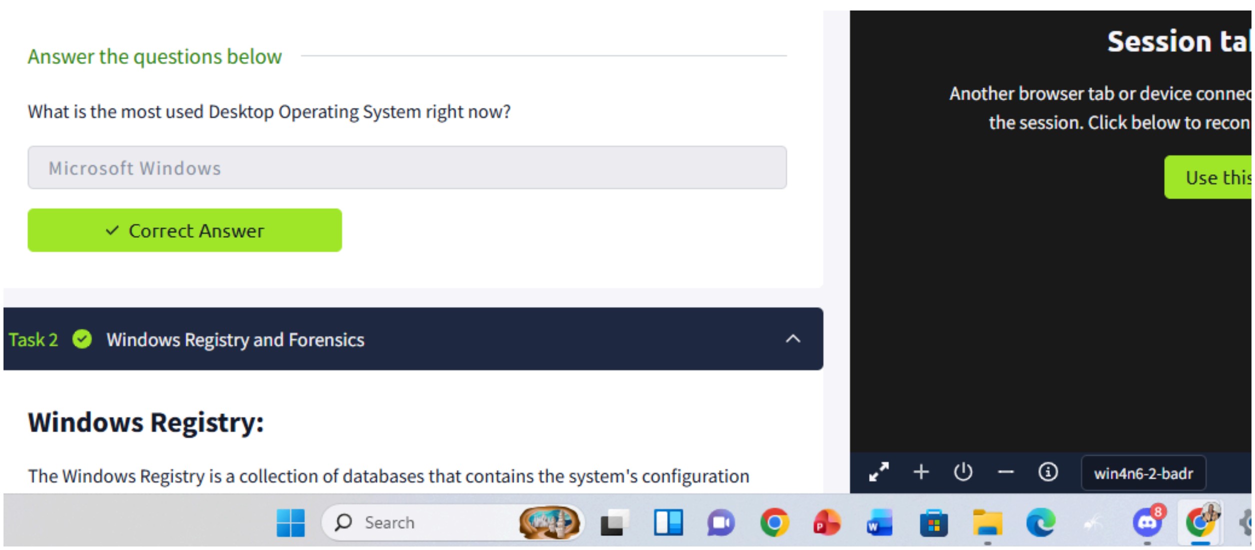
WINDOWS FORENSIC 1 WITH TRY HACK ME

1.1 Introduction

Computer forensics, a branch of digital forensics, involves collecting and analyzing evidence from computers and other digital devices to investigate activities in both legal and corporate contexts. On Windows systems, which dominate the desktop market, forensic analysis focuses on artifacts—traces of user activity stored in locations like the registry, user profiles, and application files—that can reconstruct what actions were performed. While Windows tracks user preferences and activity primarily to improve the user experience, these same traces serve as crucial evidence for investigators, as demonstrated in cases like the BTK serial killer investigation. Understanding how to identify and interpret these artifacts is essential for performing forensic analysis on Windows systems.

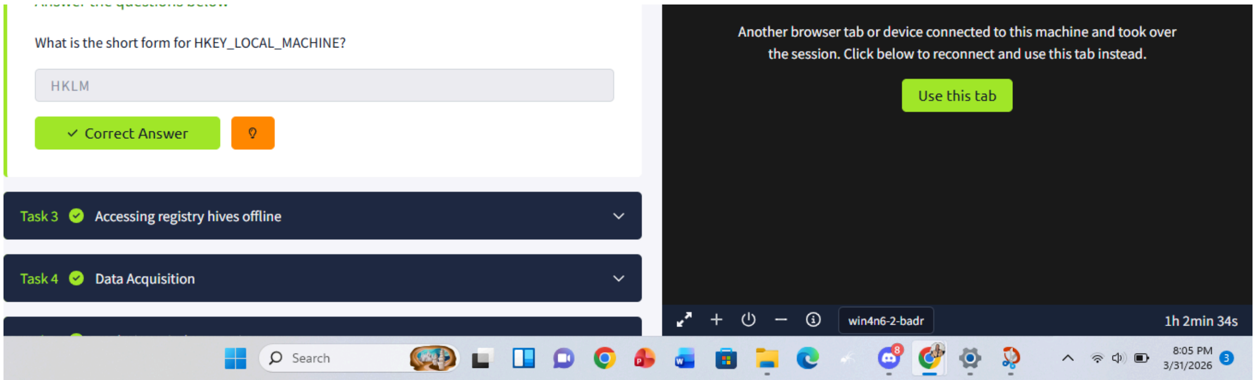
2.1 Introduction to Windows Forensic

What is the most used Desktop Operating System right now?
Answer: **Microsoft Windows**



3.1Windows Registry and Forensic

What is the short form for HKEY_LOCAL_MACHINE?
Answer: **HKLM**



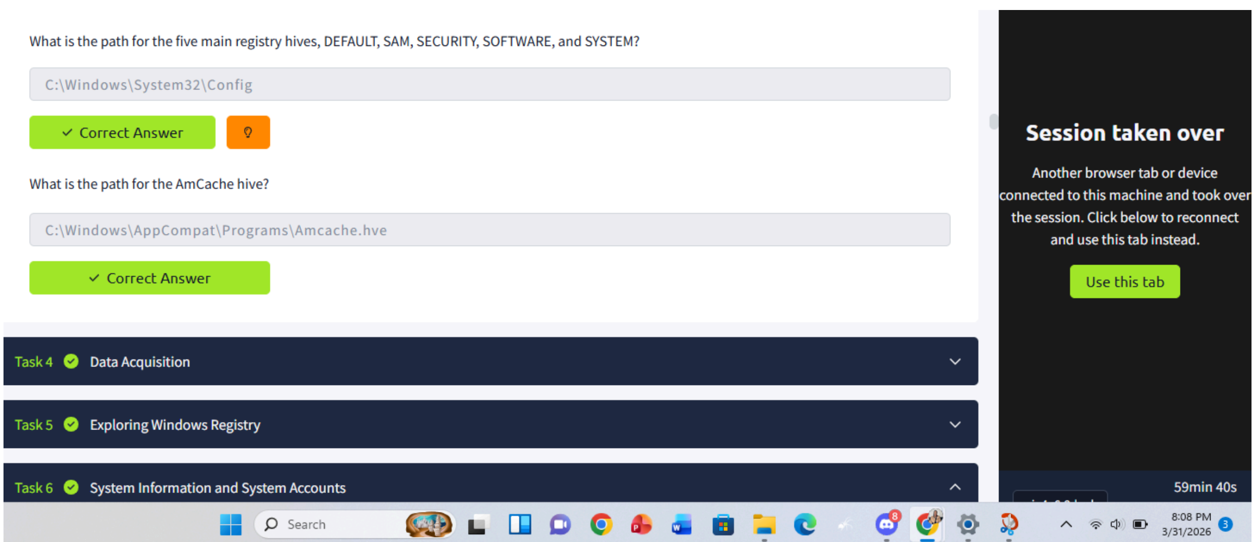
4.1Accessing Registry hive offline

a)What is the path for the five main registry hives, DEFAULT, SAM, SECURITY, SOFTWARE, and SYSTEM?

Answer:**C:\Windows\System32\Config**

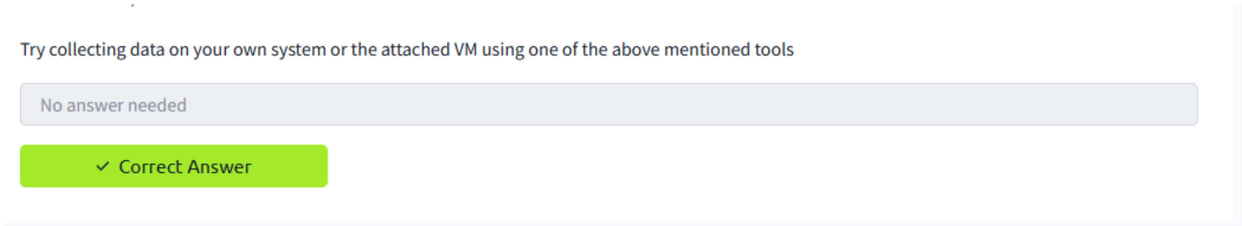
b)What is the path for the AmCache hive?

Answer:**C:\Windows\AppCompat\Programs\Amcache.hve**

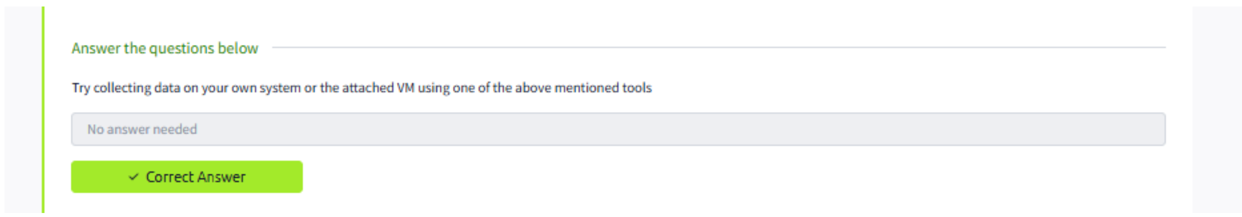


5.1 Data Acquisition

To acquire registry data for forensic analysis, investigators must bypass Windows' file restrictions on the %WINDIR%\System32\Config directory by using specialized acquisition tools rather than a simple manual copy. Tools like KAPE provide a streamlined GUI or command-line interface for live data collection, while Autopsy and FTK Imager allow for the extraction of specific hives from both live systems and disk images. While FTK Imager's "Obtain Protected Files" feature is a quick way to grab major hives from a live system, it often misses critical artifacts like Amcache.hve, making it essential to choose the right tool based on the specific evidence required for the investigation.



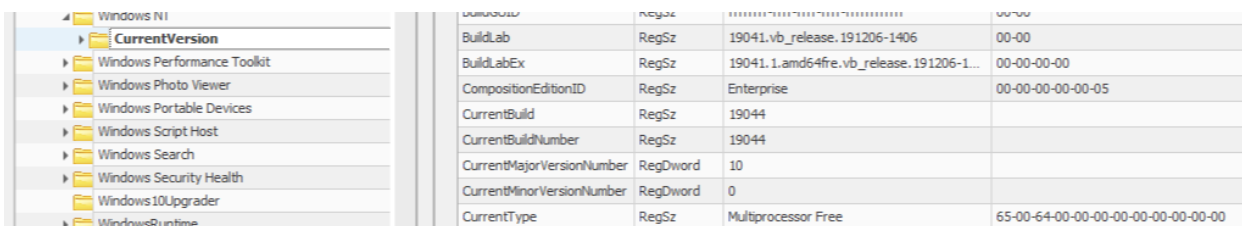
6.1 Exploring Windows Registry



7.1 System Information and system Account

a)What is the Current Build Number of the machine whose data is being investigated?

Answer:**19044**

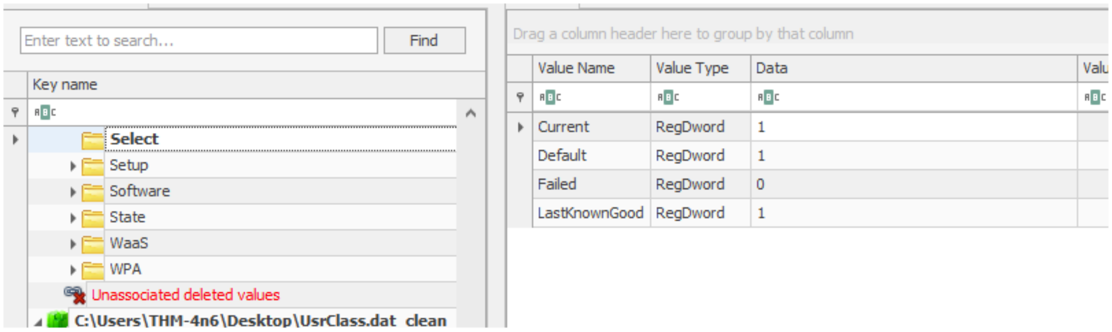


BuildLab	RegSz	19041.vb_release.191206-1406	00-00
BuildLabEx	RegSz	19041.1.amd64fre.vb_release.191206-1...	00-00-00-00
CompositionEditionID	RegSz	Enterprise	00-00-00-00-00-05
CurrentBuild	RegSz	19044	
CurrentBuildNumber	RegSz	19044	
CurrentMajorVersionNumber	RegDword	10	
CurrentMinorVersionNumber	RegDword	0	
CurrentType	RegSz	Multiprocessor Free	65-00-64-00-00-00-00-00-00-00-00-00

b)Which ControlSet contains the last known good configuration?

Answer: **1**

This is how it looks like in Registry Explorer. Take a look and answer Question # 2.



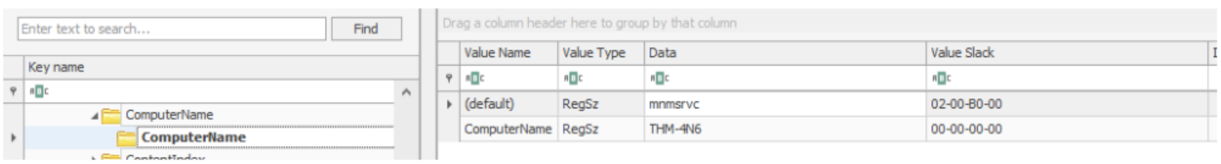
c)What is the Computer Name of the computer?

Answer:**THM-4n6**

work on. We can find the Computer Name from the following location:

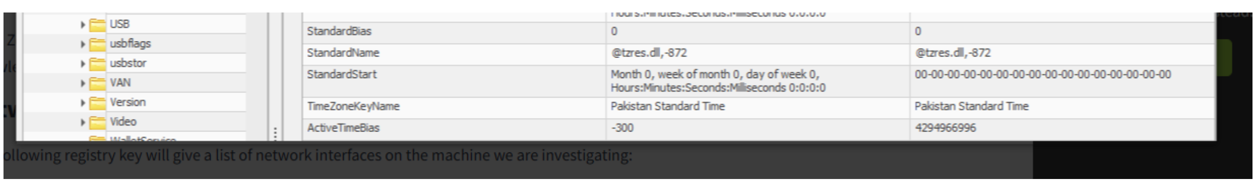
```
SYSTEM\CurrentControlSet\Control\ComputerName\ComputerName
```

Registry Explorer shows it like this. Take a look and answer Question # 3:



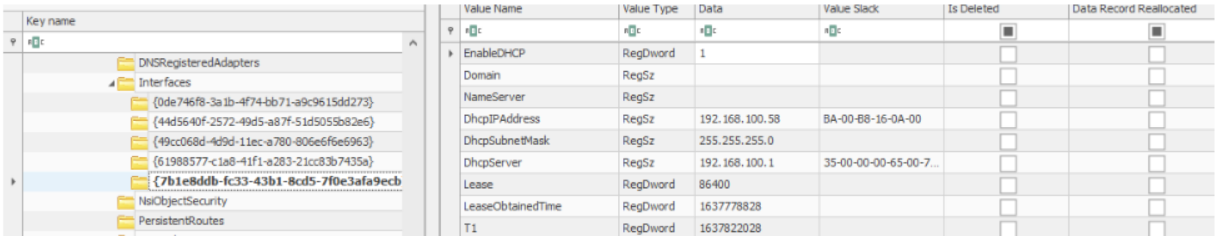
d)What is the value of the TimeZoneKeyName?

Answer:**Pakistan Standard Time**



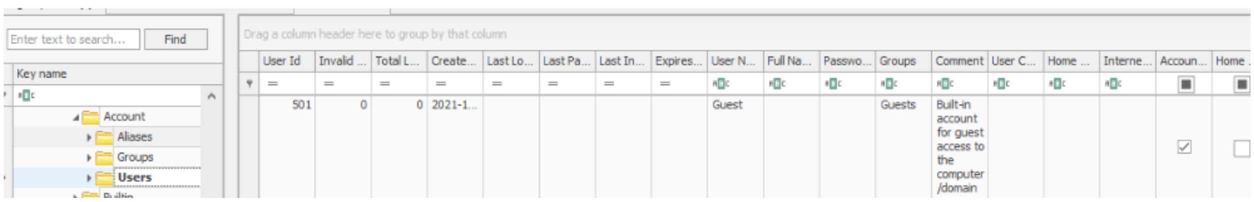
e)What is the DHCP IP address

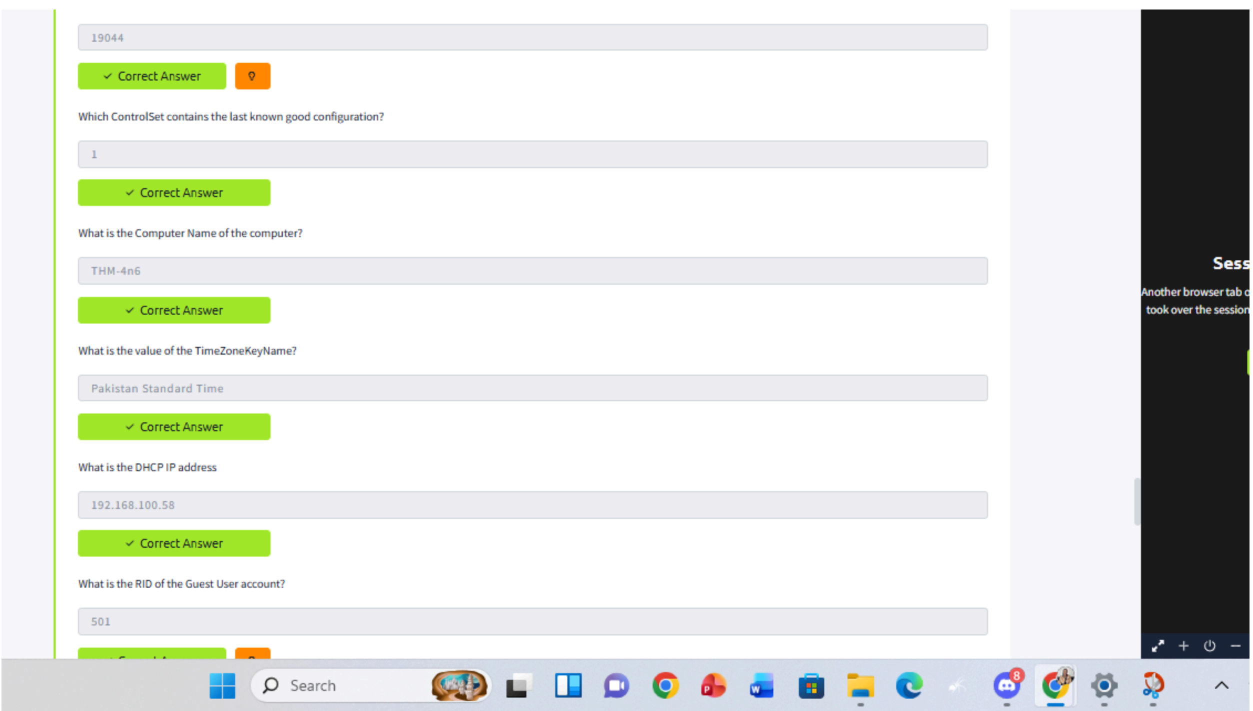
Answer:**192.168.100.58**



f)What is the RID of the Guest User account?

Answer:**501**

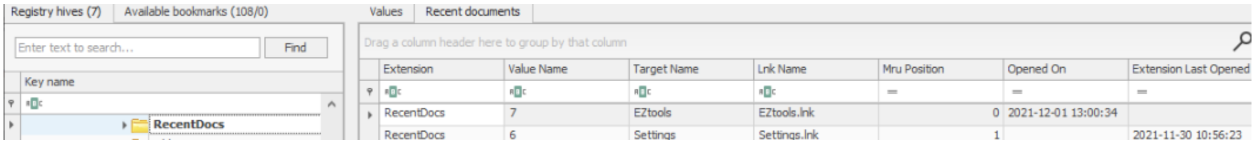




8.1 Usage or Knowledge Files /Folders

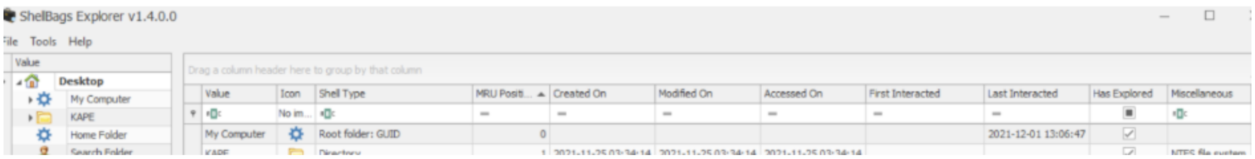
a)When were EZtools opened?

Answer:**2021-12-01 13:00:34**



b)At what time was my computer last interacted with?

Answer:**2021-12-01 13:06:47**



c)What is the Absolute Path of the file opened using notepad.exe?

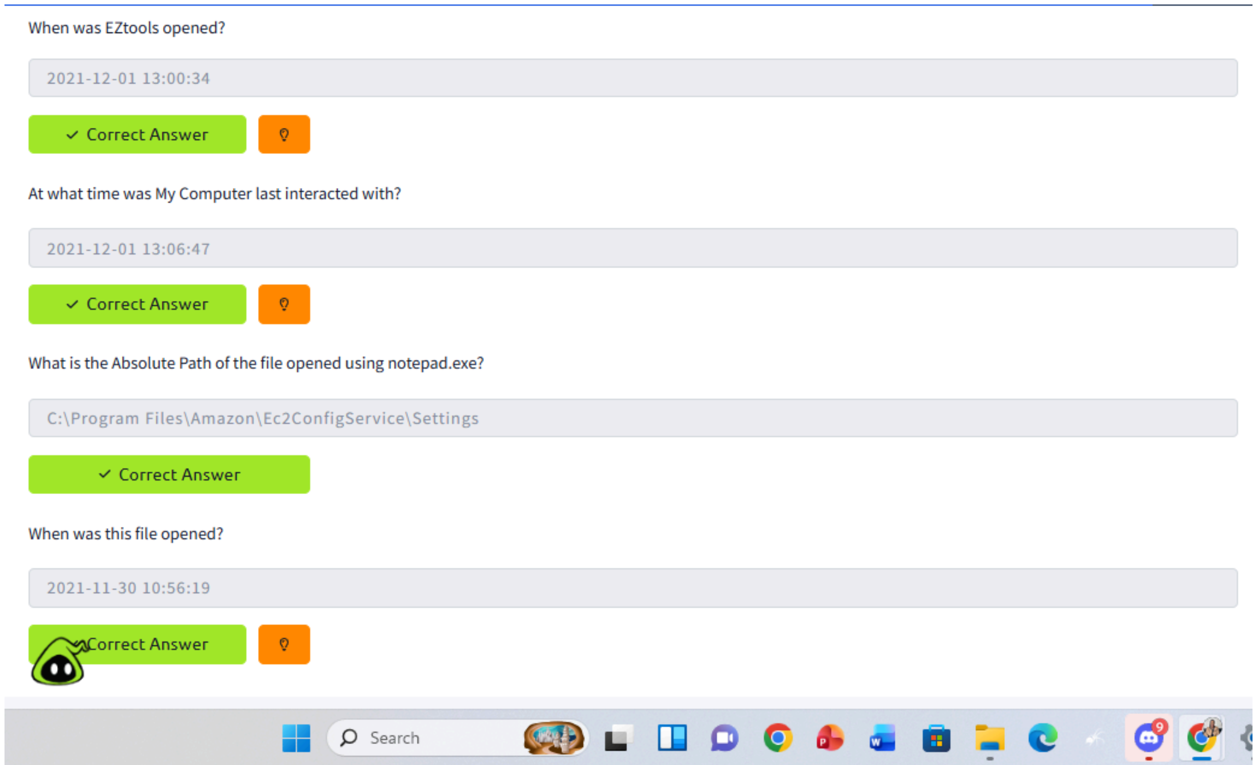
Answer:**C:\Program Files\Amazon\Ec2ConfigService\Settings**

This is how Registry Explorer shows this registry key. Take a look to answer Question # 3 and 4.

d)When was this file opened?

Answer:**2021-11-30 10:56:19**

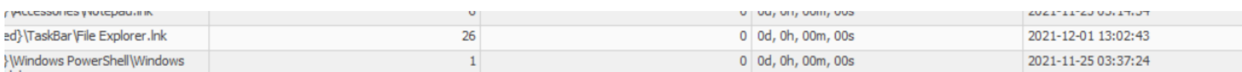
This is how Registry Explorer shows this registry key. Take a look to answer Question # 3 and 4.



9.1 Evidence of Execution

a)How many times was the File Explorer launched?

Answer: **26**



ed}\TaskBar\File Explorer.lnk	26	0	0d, 0h, 00m, 00s	2021-12-01 13:02:43
}\Windows PowerShell\Windows	1	0	0d, 0h, 00m, 00s	2021-11-25 03:37:24

b)What is another name for ShimCache?

Answer:**AppCompatCache**

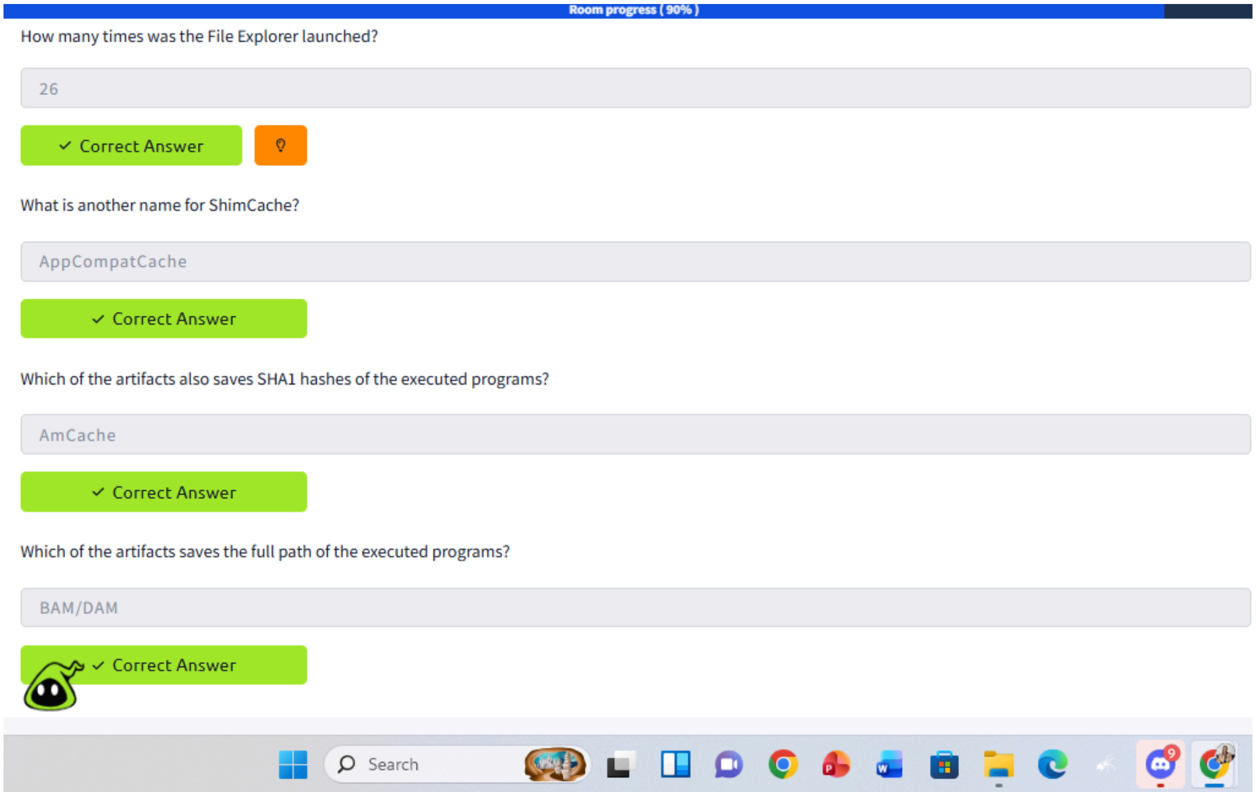
ShimCache is a mechanism used to keep track of application compatibility with the OS and tracks all applications launched on the Windows is to ensure backward compatibility of applications. It is also called Application Compatibility Cache (AppCompatCache). in the SYSTEM hive:

c)Which of the artifacts also saves SHA1 hashes of the executed programs?

Answer:**AmCache**

d)Which of the artifacts saves the full path of the executed programs?

Answer:**BAM/DAM**



10.1 External device /usb device forensics

What is the serial number of the device from the manufacturer 'Kingston'?

Answer:**1C6f654E59A3B0C179D366AE&0**

rsion	Disk Id	Serial Number	Device Name	Installed	First Installed	Last Connected	Last Removed
v_PMAP	{e251921f-4da2-11ec-a783-001a7dda7110}	1C6F654E59A3B0C179D366AE&0	Kingston DataTraveler 2.0 USB Device	2021-11-24 18:25...	2021-11-24 18:25...	2021-11-24 18:40...	

What is the name of this device?

Answer:**Kingston Data Traveler 2.0 USB Device**

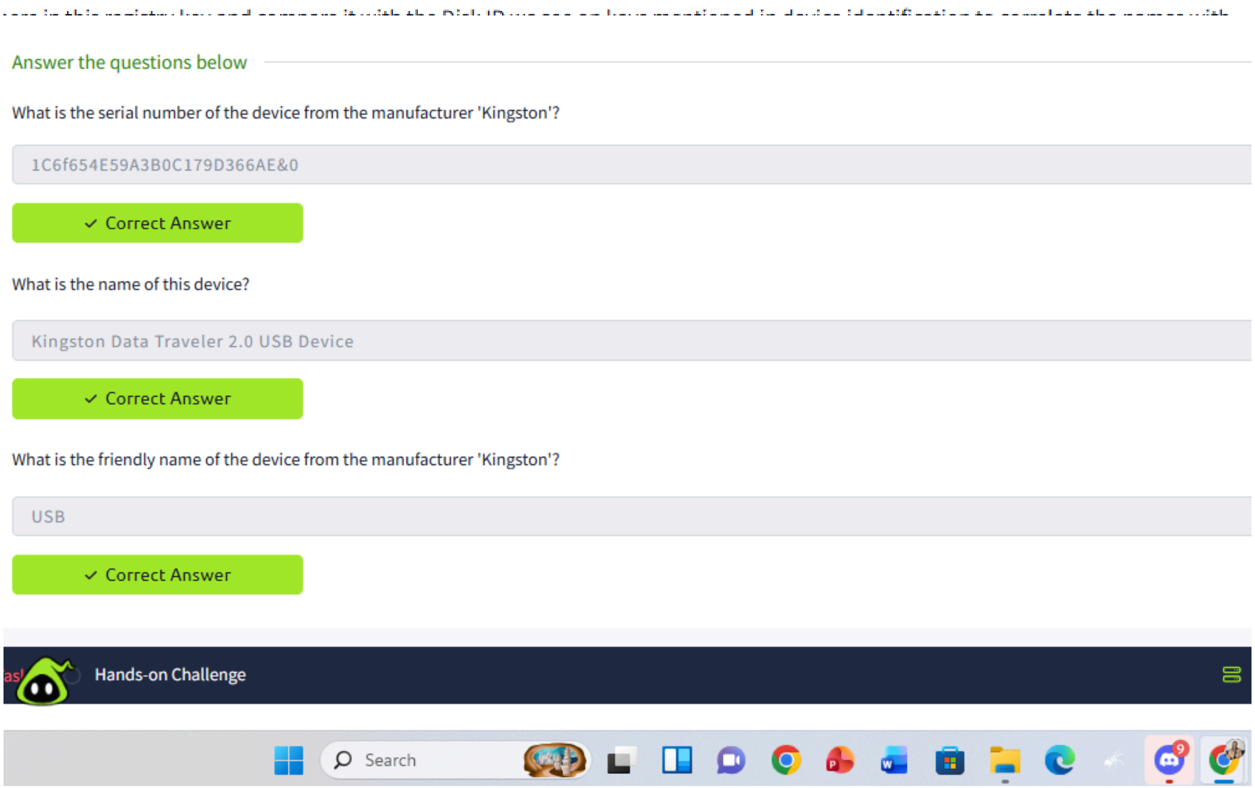
rsion	Disk Id	Serial Number	Device Name	Installed	First Installed	Last Connected	Last Removed
v_PMAP	{e251921f-4da2-11ec-a783-001a7dda7110}	1C6F654E59A3B0C179D366AE&0	Kingston DataTraveler 2.0 USB Device	2021-11-24 18:25...	2021-11-24 18:25...	2021-11-24 18:40...	

What is the friendly name of the device from the manufacturer 'Kingston'?
Answer:**USB**

alues | Windows Portable Devices

ag a column header here to group by that column

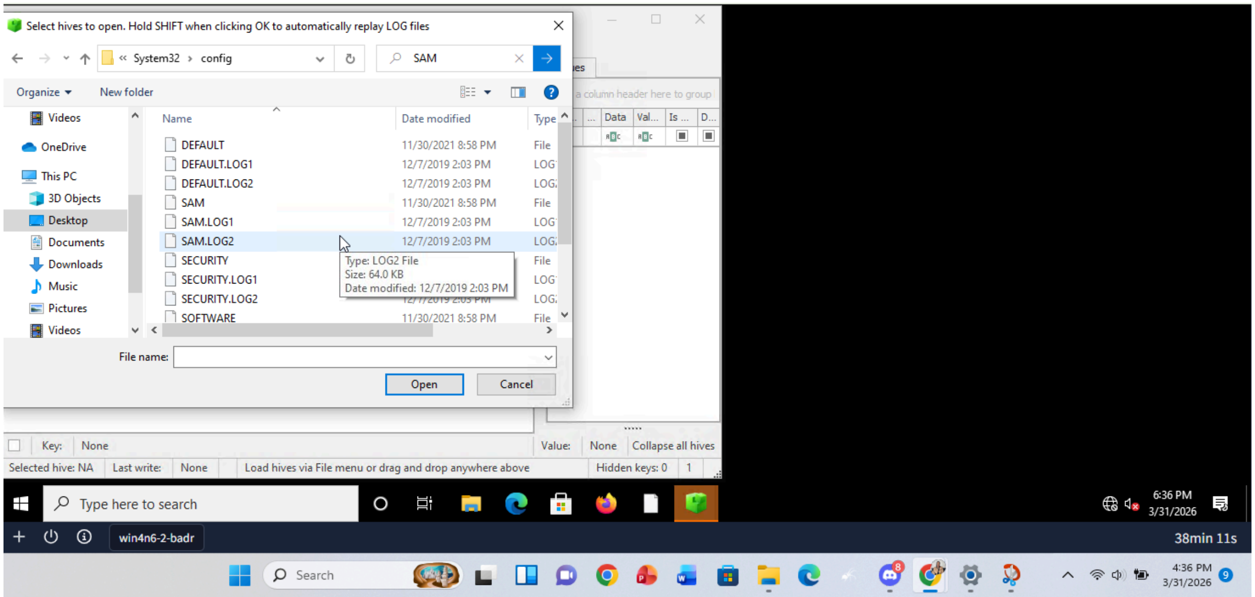
Timestamp	Device	Serial Number	Guid	Friendly Name
2021-11-25 07:16:54			{E251921F-4DA2-11EC-A783-001A7DDA7110 }	USB
2021-11-25 07:16:54			{F529A9D6-4D9E-11EC-A782-001A7DDA7110 }	New Volume



11.1 Hands on Challenge

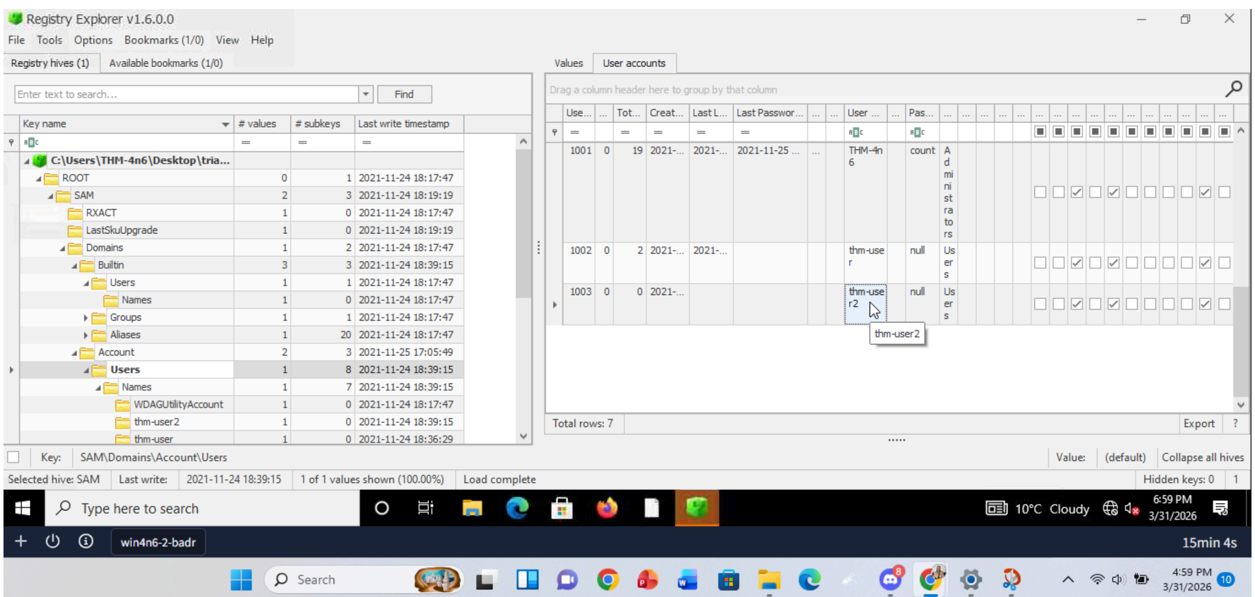
a)How many user created accounts are present on the system?

Answer: **3**



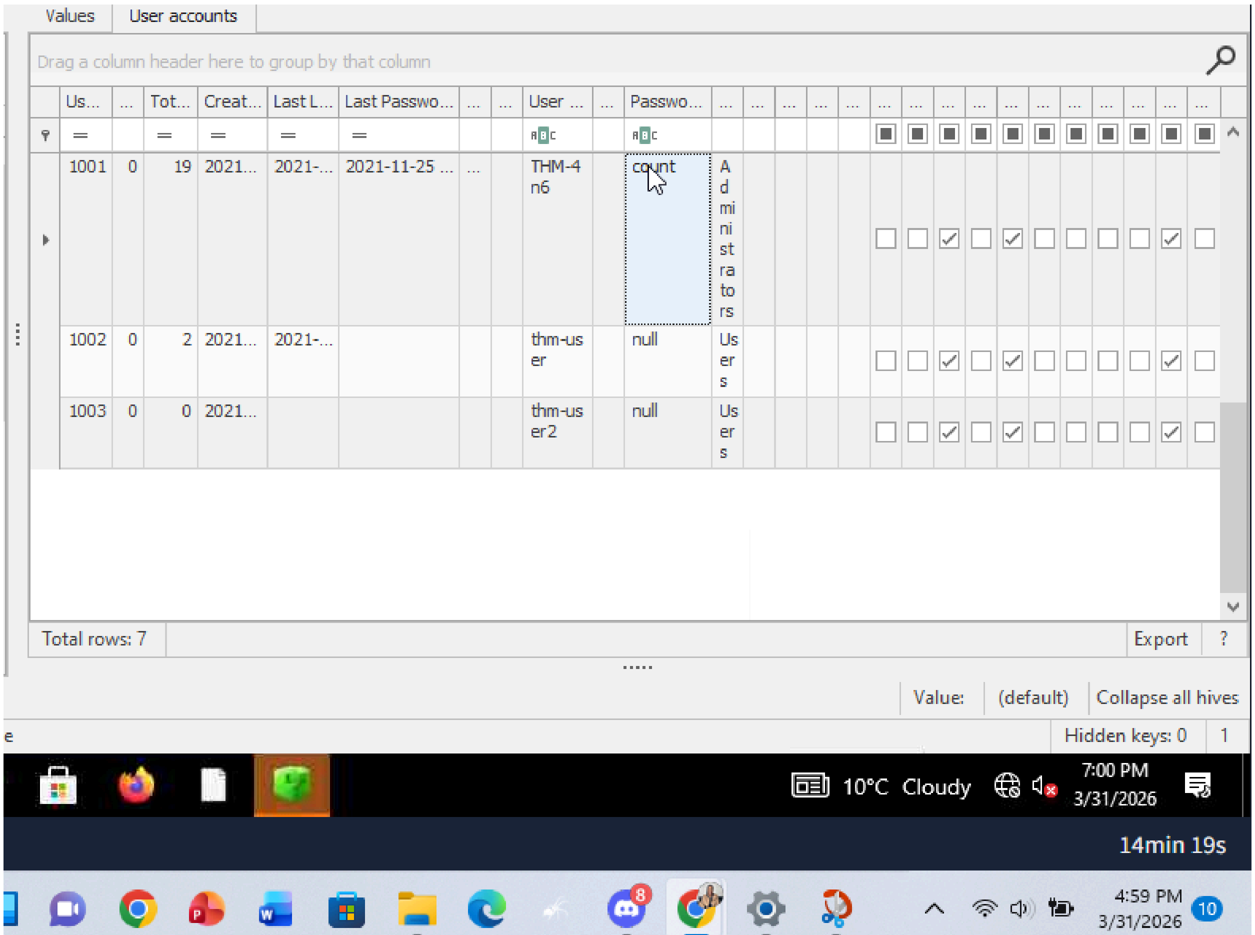
b)What is the username of the account that has never been logged in?

Answer:**thm-user2**



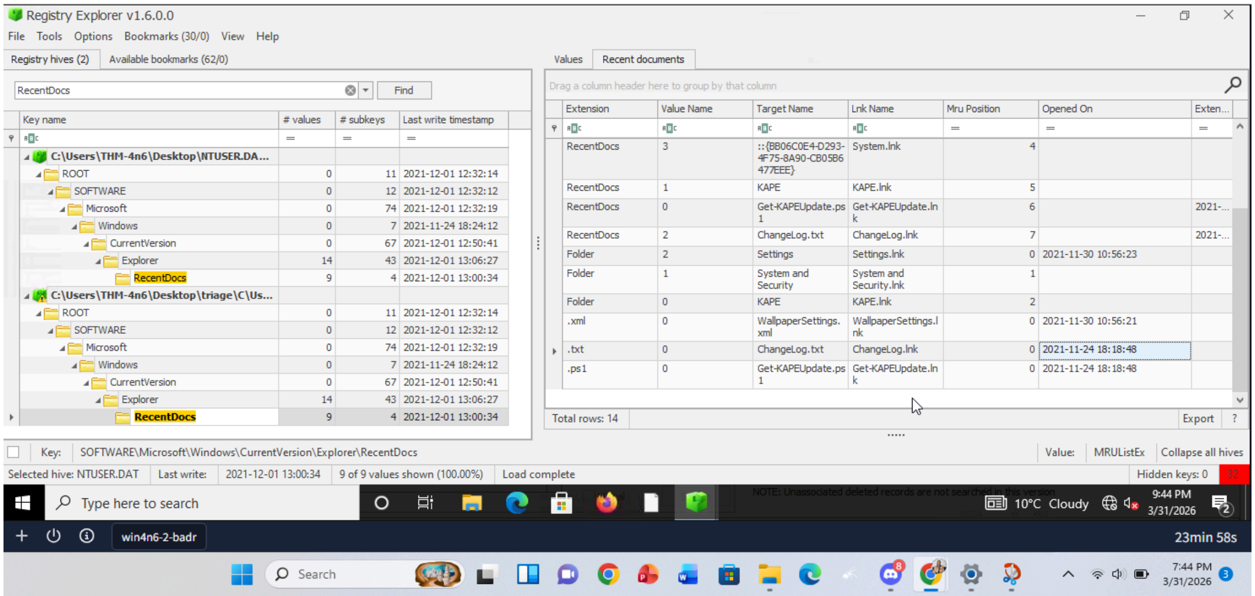
c)What's the password hint for the user THM-4n6?

Answer:**Count**



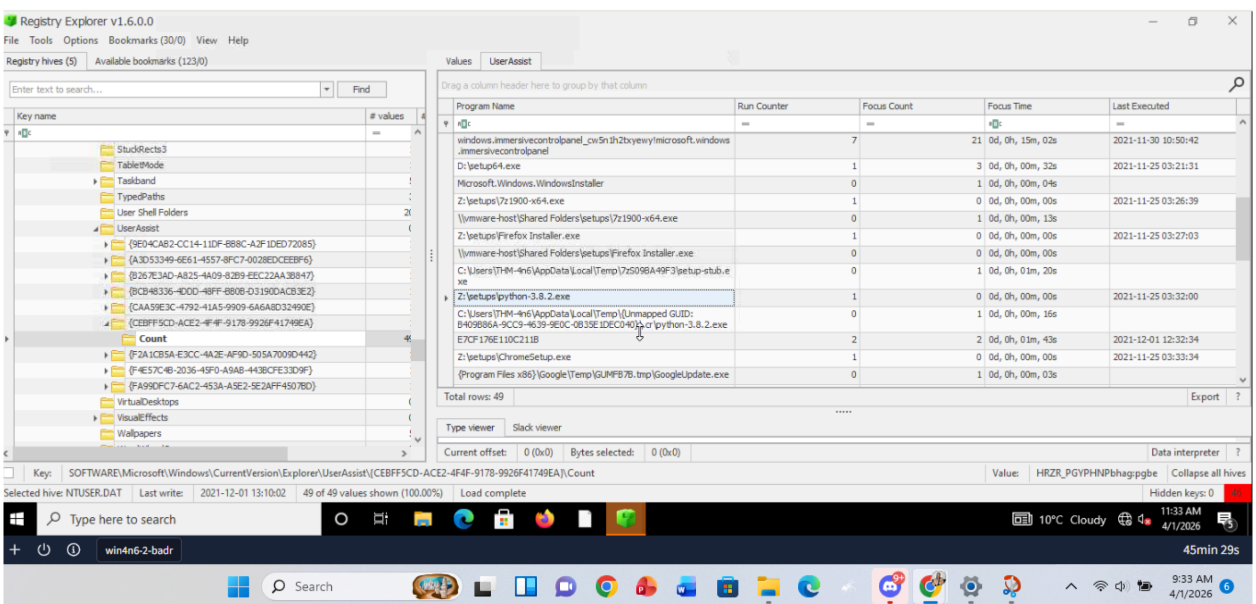
d)When was the file 'Changelog.txt' accessed?

Answer:**2021-11-24 18:18:48**



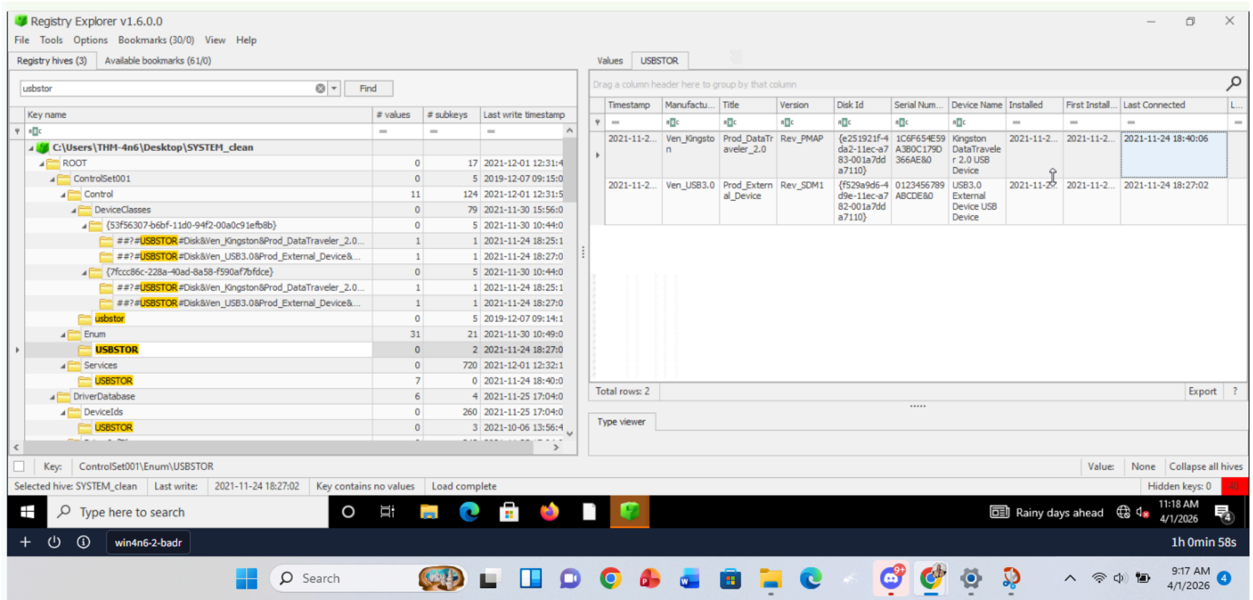
e)What is the complete path from where the python 3.8.2 installer was run?

Answer:**Z:\setups\python-3.8.2.exe**

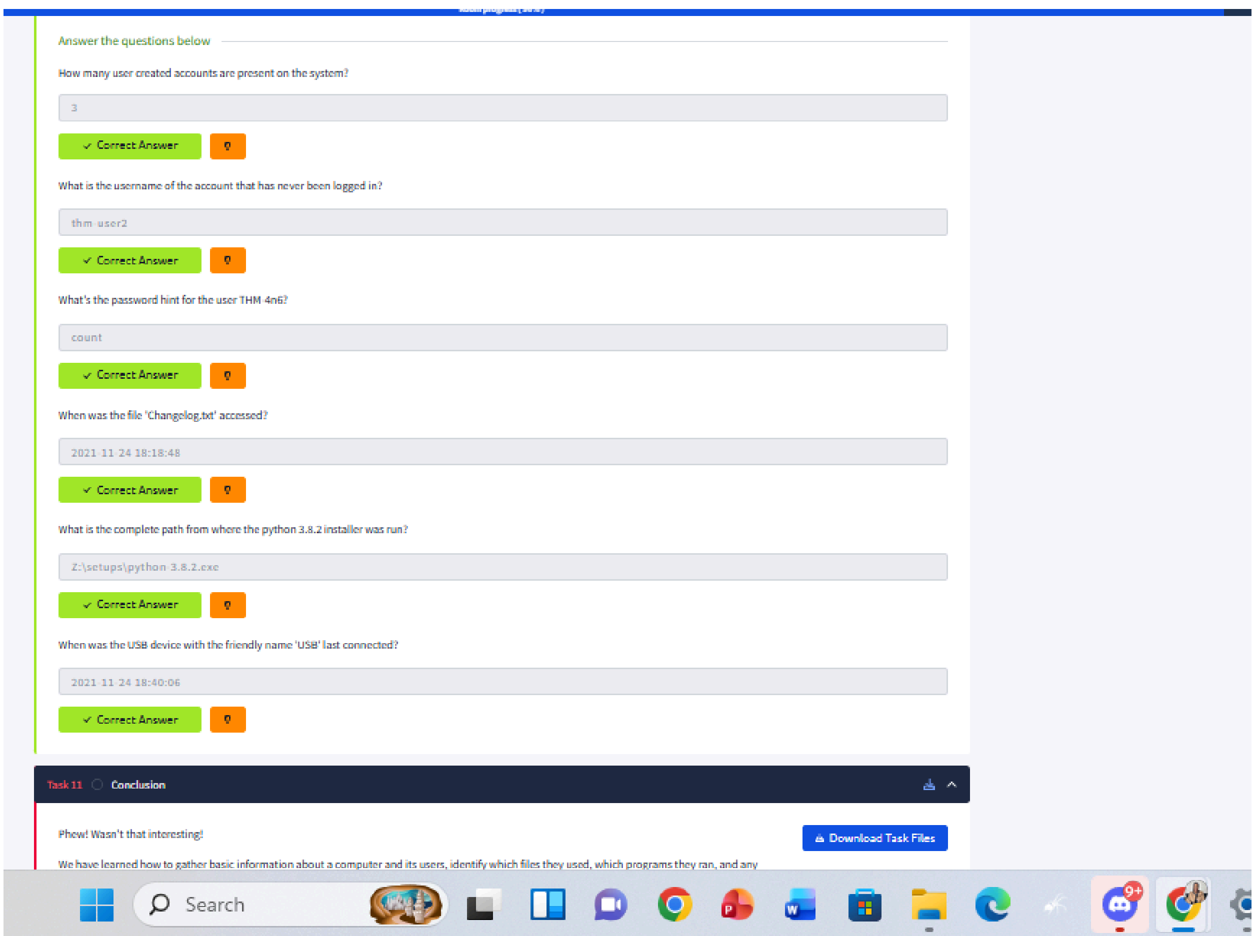


f)When was the USB device with the friendly name 'USB' last connected?

Answer:**2021-11-24 18:40:06**



Below is a screenshot to my answers



Conclusion and Summary

I understand registry forensics as the process of reconstructing a user's digital footprint by acquiring and analyzing Windows Registry "hives," such as SAM, SYSTEM, and NTUSER.DAT. Since these files are usually locked while the system is running, I would use tools like KAPE or FTK Imager to extract them safely.

After acquiring the hives, I can load them into a tool like Registry Explorer and replay transaction logs to make sure the data is complete and up to date. From there, I navigate through specific registry keys to identify useful forensic artifacts.

For instance, I can use the SAM hive to identify user account details and password hints, the SYSTEM hive to trace hardware activity like USB connections, and NTUSER.DAT to understand user behavior, such as program execution and recently accessed files.

Overall, I see the Windows Registry as a kind of "black box" that records most user and system activity, making it a very valuable source of evidence during an investigation.

Below is a screenshot and a link to the completion of the task.

https://tryhackme.com/room/windowsforensics1/congratulations?step=records